Exp No: 11	Routing at Network Layer - Static Routing

AIM:

To simulate the static routing protocol configuration using Cisco Packet Tracer.

ALGORITHM / PROCEDURE:

1. Place two routers, two switches and the PCs on the Packet Tracer workspace. Connect the PCs to the switches and the switches and routers with copper straight-through cables.
2. Assign IP addresses to the router interfaces: Router0 Gig0/0 = 192.168.1.1 and Gig0/1 = 192.168.2.1; Router1 Gig0/0 = 192.168.2.2 and Gig0/1 = 192.168.3.1 (mask 255.255.255.0). Use `no shutdown` on every interface.
3. Assign static IP addresses to the PCs and set the default gateway of each PC to the address of its router interface.
4. Ping from PC0 to the router interface 192.168.1.1. The ping fails while the interfaces are down and succeeds after the interfaces are configured.
5. To reach the networks that are not directly connected, add a static route on each router: `ip route <network> <mask> <next-hop>`.
6. Verify with `show ip route` and ping between the PCs of the different networks.

Commands used:

Purpose	Command
Enter configuration mode	enable, configure terminal
Select interface	interface gigabitEthernet 0/0
Assign IP address	ip address 192.168.1.1 255.255.255.0
Bring the interface up	no shutdown
Static route	ip route <network> <mask> <next-hop>
Check the routing table	show ip route

DIAGRAM:

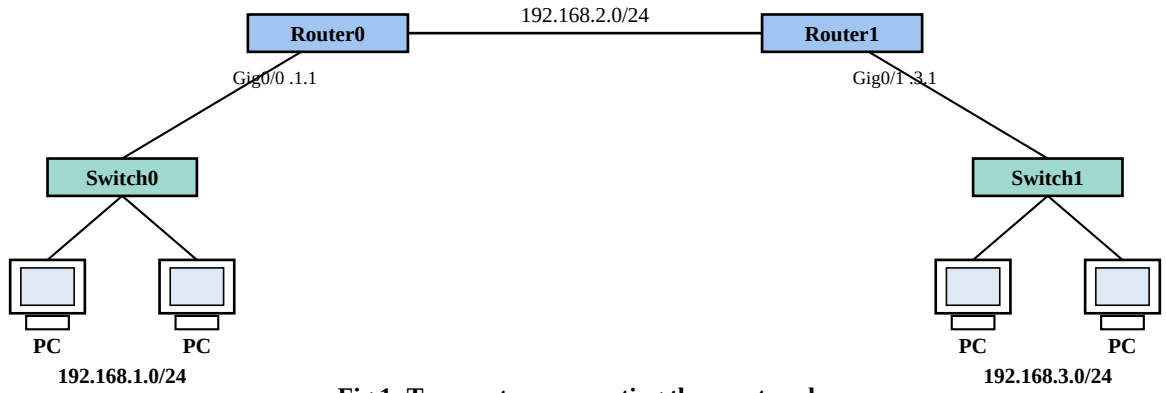
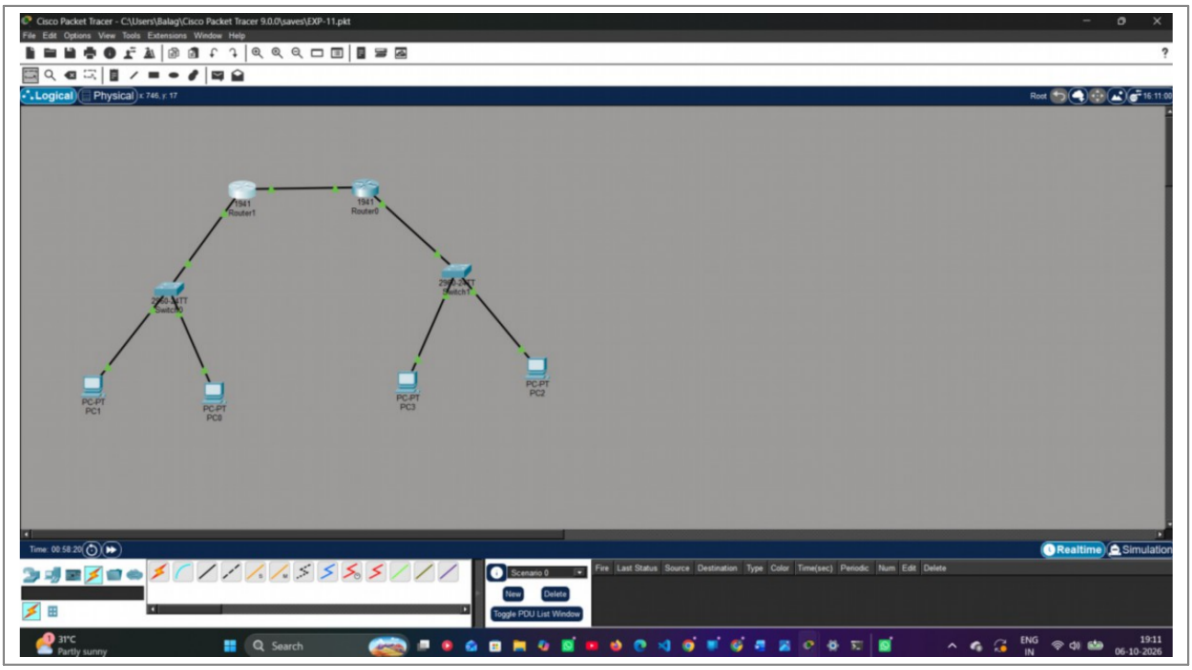
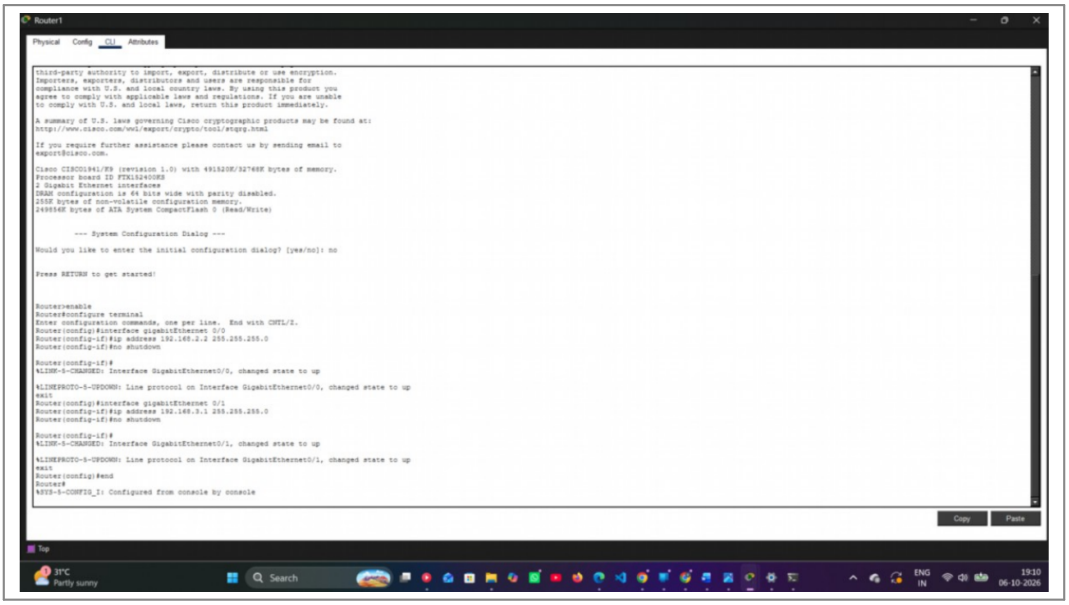


Fig 1: Two routers connecting three networks

OUTPUT:



Packet Tracer topology



Router0 - interface configuration (192.168.1.1 and 192.168.2.1)

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.2.2 255.255.255.0
Router(config-if)#no shutdown
```

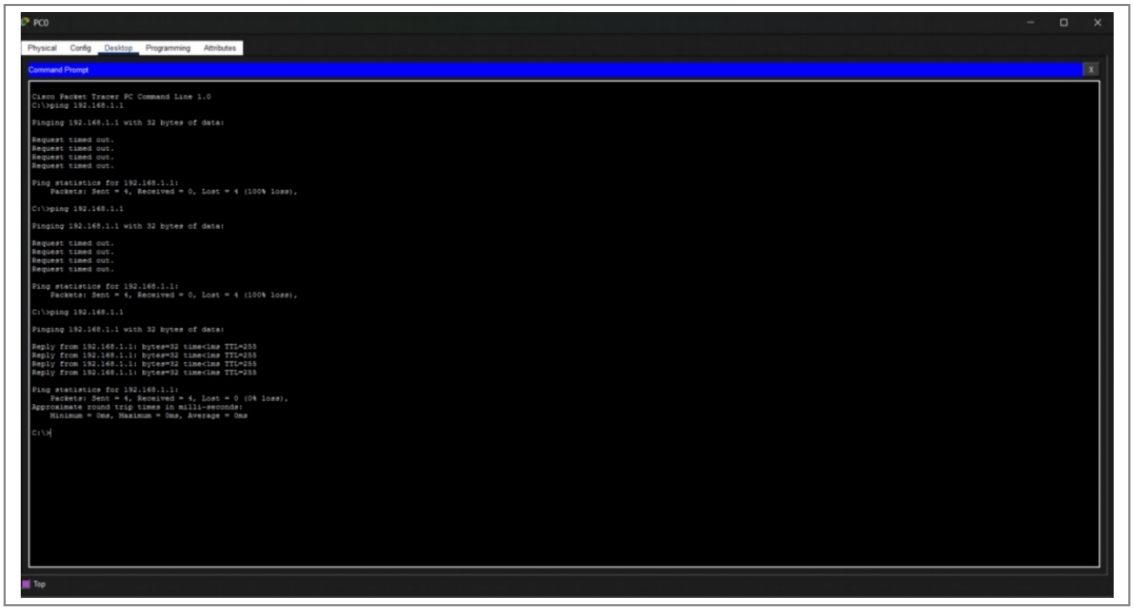
Router1 - interface configuration (192.168.2.2)

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown
```

Router1 - second interface (192.168.3.1)

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router1 - configuration saved from console



PC0 - ping 192.168.1.1: first two attempts fail (100% loss) before the interface is up, the third succeeds (4/4 replies, TTL = 255)

RESULT:

Thus the router interfaces were configured in Cisco Packet Tracer and the connectivity from the PC to the router was verified with the ping command, which failed before the interface configuration and succeeded after it.